Design, Implementation and Evaluation of Ultrasonic Measurement System using ROS and MQTT

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***Abstract*—Performing inspection tasks in industrial environment require high accuracy systems. Capabilities of this system should have potential to minimize operator involvement in hazardous areas. Currently, operation of ultrasonic thickness (UT) measurement cannot be monitored and controlled by robot pilot. One operator should be responsible to inform pilot while pilot is forcing robot to measure the thickness of surface. This article presents designing and implementation of a UT measurement system that can be visualized and controlled by users and pilot simultaneously through an interface between Robotic Operating System (ROS) and Message Queue Telemetry Transport (MQTT). MQTT is one of the data communication protocols for Internet of Things (IoT). The interface between ROS and MQTT is called ROS-MQTT bridge. Moreover, we make a comparison between ROS-MQTT bridge and client signal for each topic by using normalized cross correlation method. We used two samples of steel, with and without coating. The results show that the normalized cross-correlation of 1 for thickness numbers between two signals in client and ROS-MQTT bridge sides. The client side received thickness number with less than 1 millisecond delay. Also, it is same for another topic such as dispenser. The contribution of this work is to develop UT measurement system with other sensors such as LiDAR sensors and camera to control robot motion. The second contribution is to assist robot pilot and other users to visualize and control the UT measurement system remotely from safe area.**

***Index Terms*—UT measurement system, ROS-MQTT bridge, normalized cross correlation, time-lag**

I. Introduction

Industrial environments often contain facilities and areas that are hazardous to humans or difficult to access. The gas tank unit of a crude oil plant is one example of such an environment type as shown in Figure 1. It demonstrates such a unit, and UT measurement then present as an application example through this work. If it is required to do thickness measurement such areas, have to be taken enormous wariness to minimize a high risk for operators. Moreover, it is necessary to install scaffolding as shown in Figure 1(a). The purposed UT measurement system can be attached to any robot such as drone, magnetic wall climbing or crawler robots in order to measure thickness of surface remotely from safe area as shown in Figure 1(b). The purposed UT measurement system in this study, capable of controlling ultrasonic gel dispenser and monitoring surface thickness remotely from safe area for robot pilot and other users in order to minimize operator exposure to high risk task. However, the purposed UT measurement system has been attached to drone in this study, we only concentrate on real time data transfer for monitoring and controlling the UT probe and ultrasonic gel dispenser with MQTT and ROS. In this study, the drone is controlled manually by a pilot who is able to control and monitor the UT measurement system. This study is divided into the following sections. Section 2 explains about existing technologies for thickness measurement system and approaches of real time data transfer in IoT and its applications. Furthermore, for inter-process communication (IPC), visualization and the existing ultrasonic hardware will be studied in this section. Section 3 of this work presents purpose system architecture and integration. Also, describe signal comparison approaches in this section. Section 4 explains results with respect to the presented hardware and algorithm for thickness values. The data has been collected from two sample blocks. Also, this section describes signal comparison of gel dispenser status in ROS-MQTT bridge and client side. It shows UT measurement system was deployed on drone controller in order to measure thickness of surface and

Fig. 1. (a) A crude oil tank and scaffold. (b) UT measurement system attached to robot

assist the pilot for monitoring thickness value and controlling the ultrasonic gel dispenser. Furthermore, it shows purposed UT measurement system is integrated with vision system in ROS. Despite the fact that the proposed UT system and sensor tests were conducted inside our robotic laboratory and PETRONAS Research Center, we believe that the importance of our solution and results extend to a much wide class of scenarios with different robots. With the proposed UT system for pilot assistance, we successfully demonstrated that measurement of thickness for metal and coated metal material with high accuracy.

II. PROBLEM STATEMENT AND LITERATURE REVIEW

One of the existing technologies in the market is the Tritex drone thickness measurement as shown in Figure 2. This technology is able to transmit measurement remotely to a PC or laptop up to 500 meters away in real time [1]. One of the problems of this technology is that only one user is able to monitor and control the UT measurement system. The second problem is that the Tritex software cannot be integrated with other sensors such as camera and LiDAR sensors in order to control robot motion. We propose integration of MQTT communication protocol and ROS software to solve the issue. Thus, applications and concept of MQTT as a communication protocol for IoT and ROS are investigated in this section.

IoT model designs the connection of sensors through networks that allows for controlling their performance and distribution data in real time. One of application of this IoT model applies for home automation in which it inspires from computer science, automation and new communication technologies [2,3]. All applications aimed to improve control and monitoring data of sensors [3]. IoT's challenges are to bridge physical world and the information world. For instance, the processing of data acquires from electronic devices by using an interface among users and the devices [2,3]. Sensors obtain physical data such as thickness of surface or distance from object and transmit to server in order to visualize on the application interface so that users could make decision to active actuators [4]. Some IoT technologies have been designed for communication among the sensors and actuators installed in industry, either by independent wired structure [5,6]. Some of the latest technologies have been utilized Wireless Sensor Networks (WSNs), which successfully implemented to fields like industry [7-10], public safety [11,12], telemetry [13,14], or train monitoring [15]. IoT communication protocol is MQTT means that message queue telemetry transport. It is messaging protocol. It is messaging protocol or data transfer protocol and is popular because it is very lightweight. Therefore, it requires very less amount of data to transfer and low power. It is preferred for low power application and low-power microcontrollers such as NodeMCU. Therefore, lightweight and low power protocol make it very popular in the IoT field [4]. In other hand, controlling robot motion require other sensors to be visualize from robotic operating software (ROS).

Robotic systems have been developing since 2003 by the Active Vision Group (AGAS). In the RoboCup1 competitions had been tested the developed software on various robots. Robotic Operating System software allows user to design complex software [16-17]. It provides a method in which a network of processes (nodes) connect to central hub and they can be run from multiple devices. The main methods of creating the network are defining publisher and subscriber connections with other nodes or providing request-able services in which nodes communicate through a specific message type [18]. In next section, the integration of ROS and MQTT will be explained. Moreover, a method will be explained to compare signals in order to find the time-lag among our integration system, receiver and sender.

III. ULTRASONIC THICKNESS SYSTEM DESIGN AND IMPLEMENTATION

This section includes two parts, explaining of interface between ROS and MQTT and describing methods of signal comparison. The hardware parts are listed as follow: Tritex MG6000 (Gel Dispenser, UT probe unit), NodeMCU, OLED display and push button switch as shown in Figure 3.

A. Integration of ROS and MQTT

ROS-MQTT Bridge allows to subscribe to MQTT topics (Thickness and Dispenser) and publish contents of MQTT messages to ROS as shown in Figure 3. It is also possible to publish to MQTT from ROS topics (Dispenser and Electromagnet ON/OFF). Thickness number can be visualized on OLED display for pilot assistance and in Rviz for user. The Gel dispenser can be controlled from push button switch and GUI by pilot and user respectively. The built in led of NodeMCU8266 has been used to indicate the gel dispenser is on or off. Dependencies packages are included paho-mqtt and ROS that must be installed in order to successfully run purpose ROS-MQTT bridge. MQTT broker is Mosquitto has been run on Ubuntu 18.04.

For the sake of simplicity of activation gel dispenser process, Figure 3 can be broken down into Figure 4. When user press the push button switch, bridge subscribe to dispenser topic and get feedback from Xbee in order to publish dispenser status and client side subscribe to dispenser status and led in client side must be light on for one second. Consequently, client side (pilot) will be sure that the gel dispenser has been activated.

Data in section 4 has been collected from two test blocks as shown in Fig 5. Thickness values in Fig 5.a are 12.5mm and in Fig 5.b is variable between 17.5- 19mm.

B. A Method for Comparison between Signals

One of methods of signal comparison is correlation. It is a measure of similarity of signals. This technique is popular and had been applied in speech recognition, LiDAR sensors. There are a lot of methods for correlation measurement and Equation (1) demonstrates mathematics calculation of correlation in

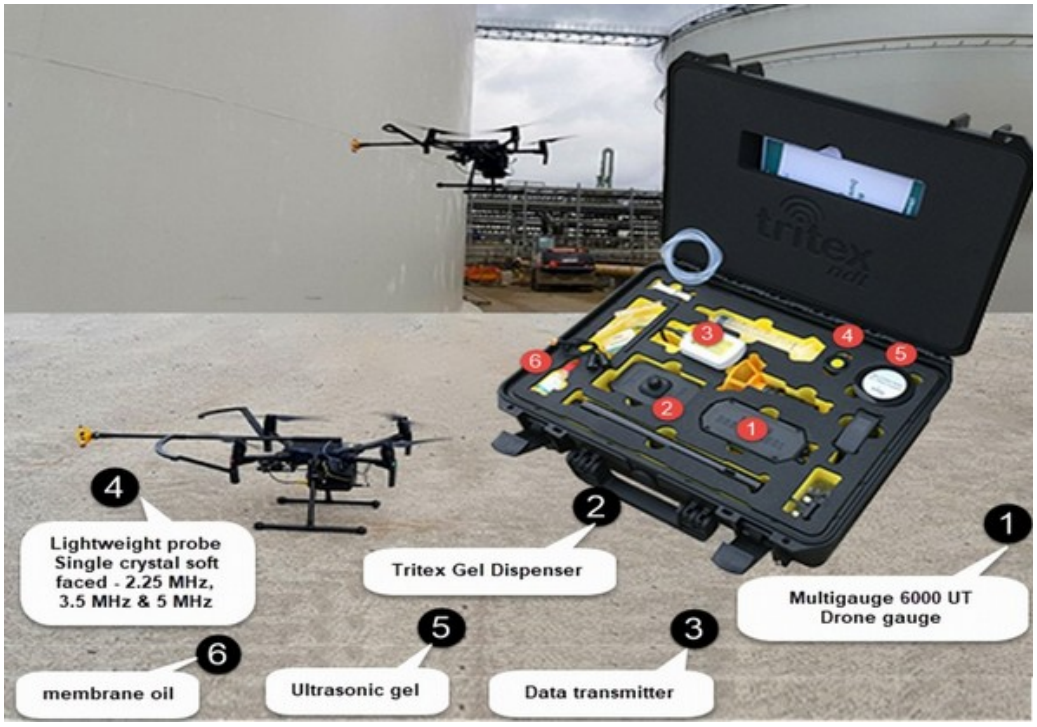

Fig. 2. Tritex multigauge 6000 drone

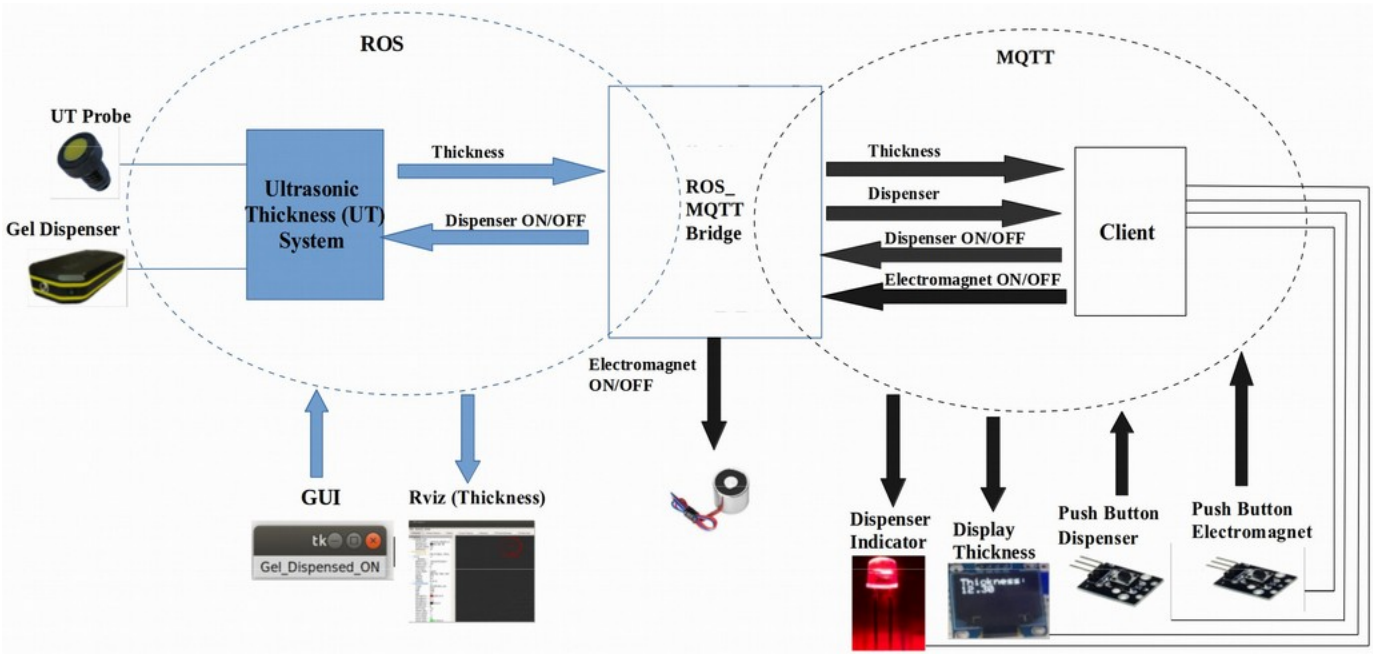

Fig. 3. Purpose system architecture

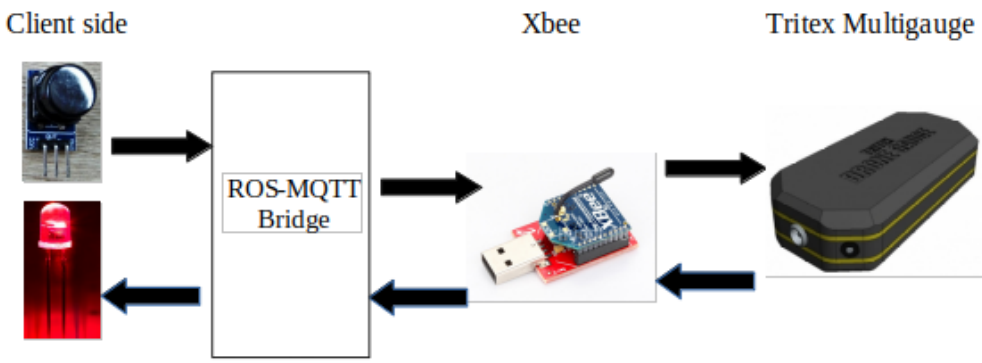

Fig. 4. Process of gel dispenser activation

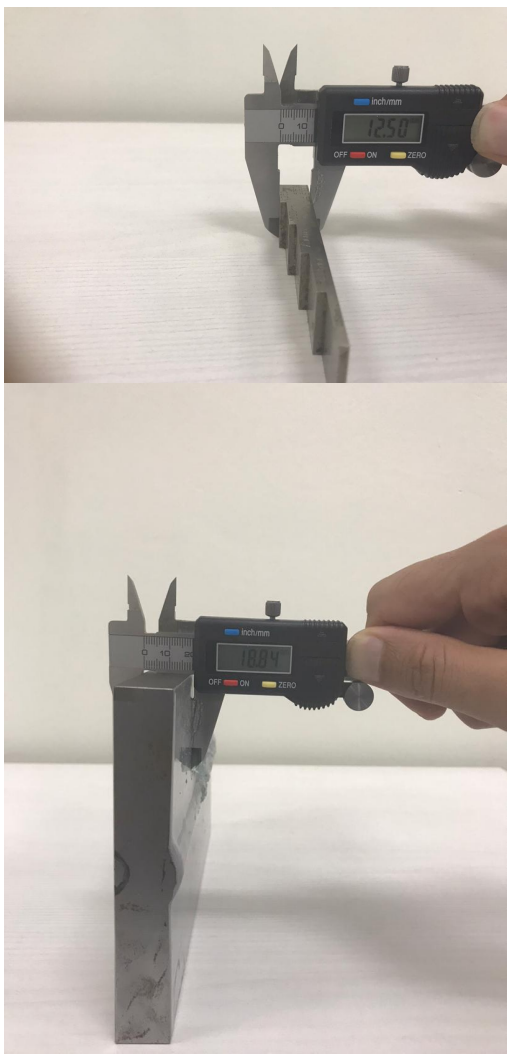

Fig. 5. (a) Without coating (b) With coating

which n is sample number zero until the sample N-1 that presents the length of signals.

$$corrnorm_{x,y} = \sum_{n=0}^{N-1} [n]y[n] \tag{1}$$

The larger the value of correlation measurement is, the more similarity there is between two signals. However, there is a problem with this particular approach. The problem is highlighted by considering following situation: If one of signal has one big energy level, the Equation (1) gives misleading result. This problem is solved by using normalized correlation rather than use basic cross correlation Equation (1). Normalize cross correlation is to scale overall of the standard cross correlation function value by the factors that related to energy level of two signals. Therefore, in next section, normalize cross correlation is applied to compare bridge and client signals for two steel samples for thickness and gel dispenser topics. Also, it can be realized that there is no time-lag between two signals when normalize correlation value is one.

$$corr_{normx,y} = \sum_{n=0}^{N-1} [n]y[n] / \sqrt{(\sum_{n=0}^{N-1} x^2[n])} * \sqrt{(\sum_{n=0}^{N-1} y^2[n])} \tag{2}$$

IV. Experiments and Results

Fig. 6 shows thickness numbers with respect to time in bridge and client sides for metal without coating and normalize correlation between bridge thickness and client thickness is one. It means that the bridge signal is as same as the client signal and there is no delay between receiving signal in the client side.

$$corr_{normB,C} = 23520.2703/23520.2703 = 1 \tag{3}$$

Fig. 7 demonstrates thickness numbers respect to time in bridge and client sides for coated metal and normalize correlation between bridge thickness and client thickness is one. It means that the bridge signal is as same as the client signal and there is not delay between receiving signal in the client side.

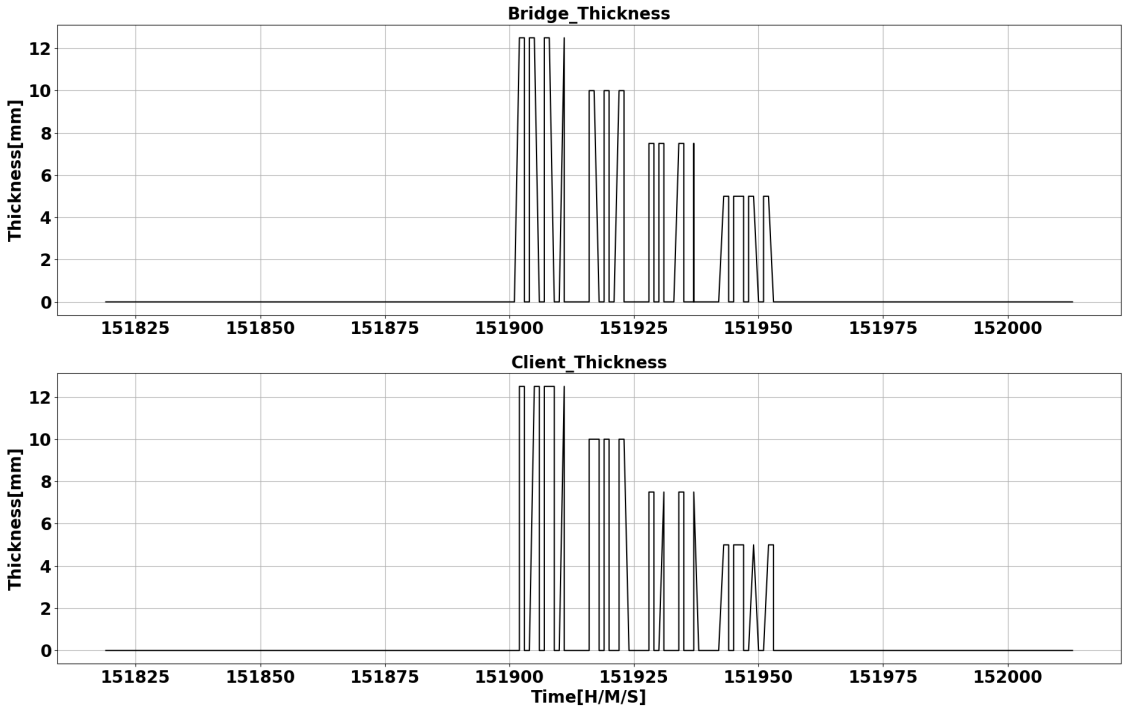

Fig. 6. Bridge and Client for thickness number of metals without coated

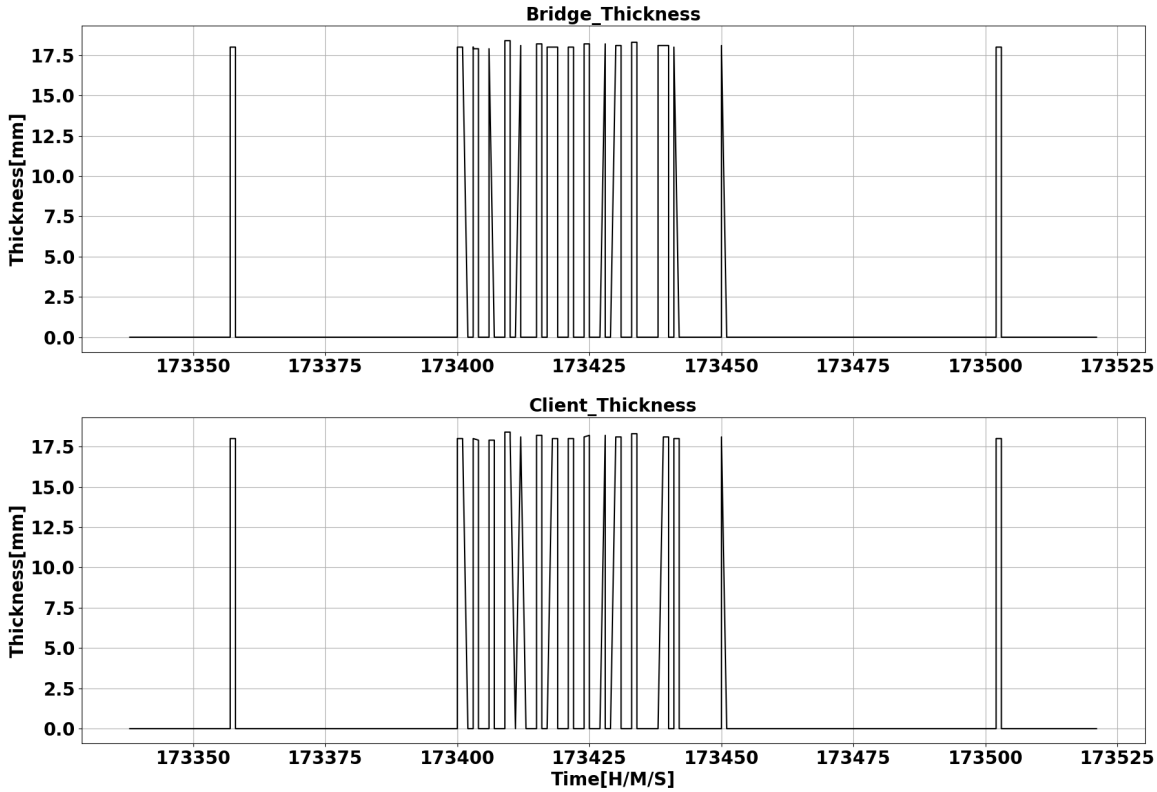

Fig. 7. Bridge and Client for thickness number of coated metals

Fig. 8 demonstrates gel dispenser status respect to time in bridge and client sides f and normalize cross correlation between bridge and client gel dispenser status is one. It means that the bridge signal is as same as the client signal and there is no delay between receiving signal in the client side.

$$corr_{norm\,B,C} = 23520.2703/23520.2703 = 1 \quad (4)$$

Fig. 9 shows integration of UT measurement system and vision system in Rviz. User can visualize the target point for measuring the thickness value. In this case, deep learning algorithm [19] can be applied to detect the target point.

Fig. 10 shows UT measurement system that was deployed on drone controller in order to assist the pilot for monitoring thickness value and controlling the ultrasonic gel dispenser.

V. CONCLUSION

This article presented an interface between ROS and MQTT in order to integrate UT measurement system with other sensors such as LiDAR and camera. Moreover, the pilot and other users are able to visualize thickness number and control the gel dispenser. From the results presented in this paper, we can note that normalized cross correlation of two signals in client and ROS-MQTT bridge sides for thickness number has not lost and client receives thickness numbers without any time-lag in term of second for both samples. Also, it is same for another topic such as dispenser. However, in this research only concentrates UT measurement system. Hence, further studies are required for integration the purposed system with LiDAR sensors as well as to control robot motion.

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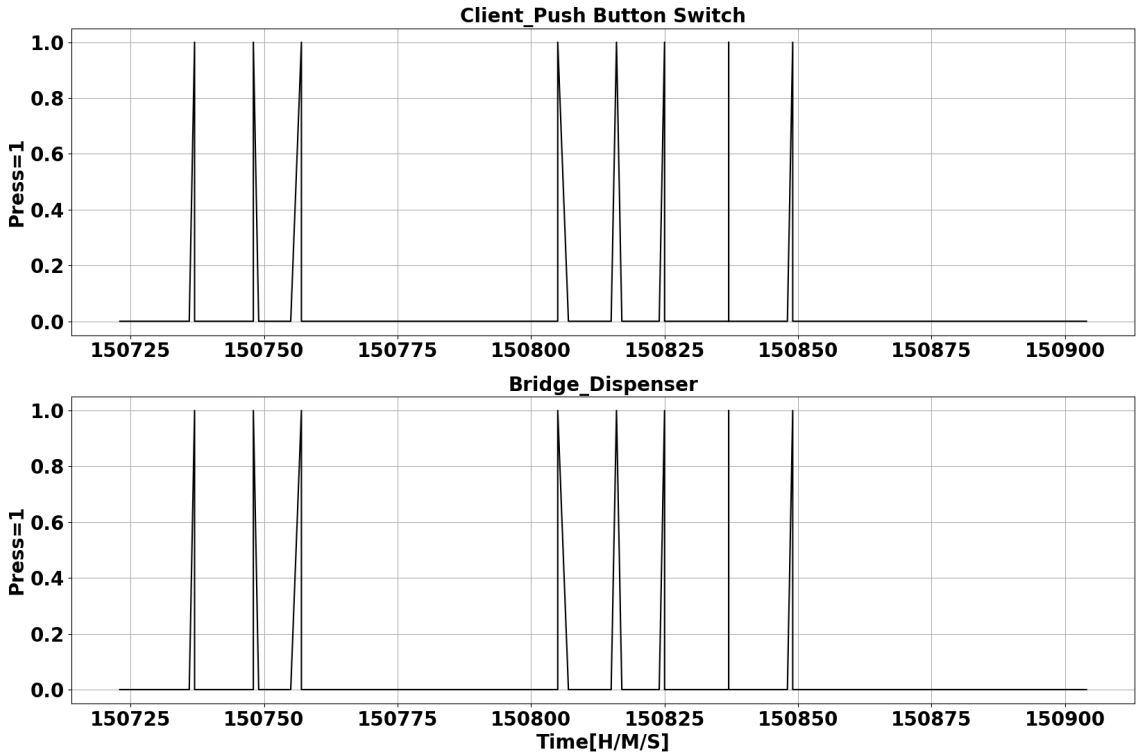

Fig. 8. Bridge and Client for activation of gel dispenser

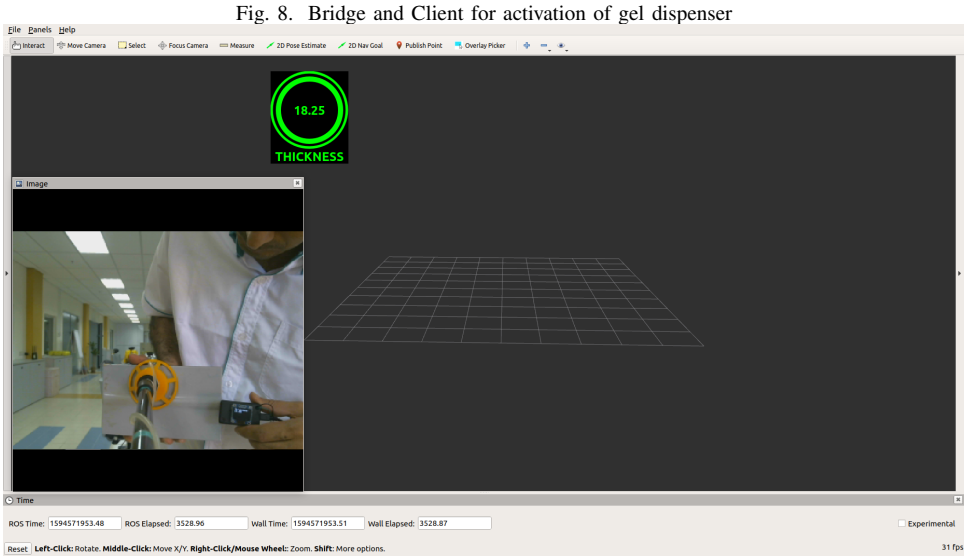

Fig. 9. Integration of thickness measurement system with camera

Fig. 10. pilot assistance system